

The Utility Problem

As electricity distribution grids are undergoing modernization, new operational measurement data streams are being created in the grid on a daily basis through the integration of various time series generation devices. This time series data has great potential for insight into generation and situational awareness. However, when a large amount of time-series data is collected, it is not possible to go through all the data manually. Therefore, it is beneficial to automatically find or flag the values that are interesting or otherwise important. One of the adapted techniques is setting the thresholds for which only desired data will be highlighted.

Setting thresholds for displaying the selected variables is very important, as it might indicate possible problems arising in the grid. Without the possibility of setting these thresholds, it is very difficult to catch these arising problems and, therefore, to solve them. The setting of the monitored units, as well as their thresholds, should be set individually (using a different methodology) and adjusted over time if necessary. This cannot be done for limits that are fixed, so a user-friendly environment must be created for these threshold changes.

For these reasons, it is beneficial for the utility to have a system in place which will allow for:

- Ingestion of different types of time series data in terms of:
 - Granularity;
 - Cadency or frequency;
 - Measured metrics.
- Setting of different metric thresholds for different types of time series data;
- Setting different levels of warning or alerts once these thresholds are breached.

With such a system in place, the utility is able to gain situational awareness, identify the issues as soon as they arise and conduct corrective measures to prevent unnecessary damages or harmful influences in the grid.

The Data Solutions and Methods

It is obvious that for this kind of system, it is necessary to have time series data on which the thresholds will be applied. The Awesense Energy Transition Platform, which can deal with large amounts of time series data and monitor this data on scale, is a great tool for this application. The differentiation of the Awesense platform is its ability to ingest disparate sources' time series data: syncing and solving timestamp granularity, updating frequency/cadency and ingesting any type of measured metrics.

Once the time series are ingested and/or integrated with the Awesense Platform, it is possible to set threshold values for any measured quantities - e.g. current, voltage, power factor etc., but also for parameters about the time-series-generating device (sensor) - e.g. communication problem, battery status etc. It is also possible to display not only customer consumption measurement parameters - but also transformer overload, PV plant threshold exceeded and

Grid Alerts and Warnings

[UC18-01] - Awesense Energy Transition Platform Use Case



other parameters. Everything can be defined very simply by selecting from a list of parameters and their threshold values. If the time series integration is implemented in a near real-time fashion, the Alerts can also be displayed in near real time. With the Awesense Energy Transition Platform, it is very easy to deploy Grid Alerts in the entire grid.

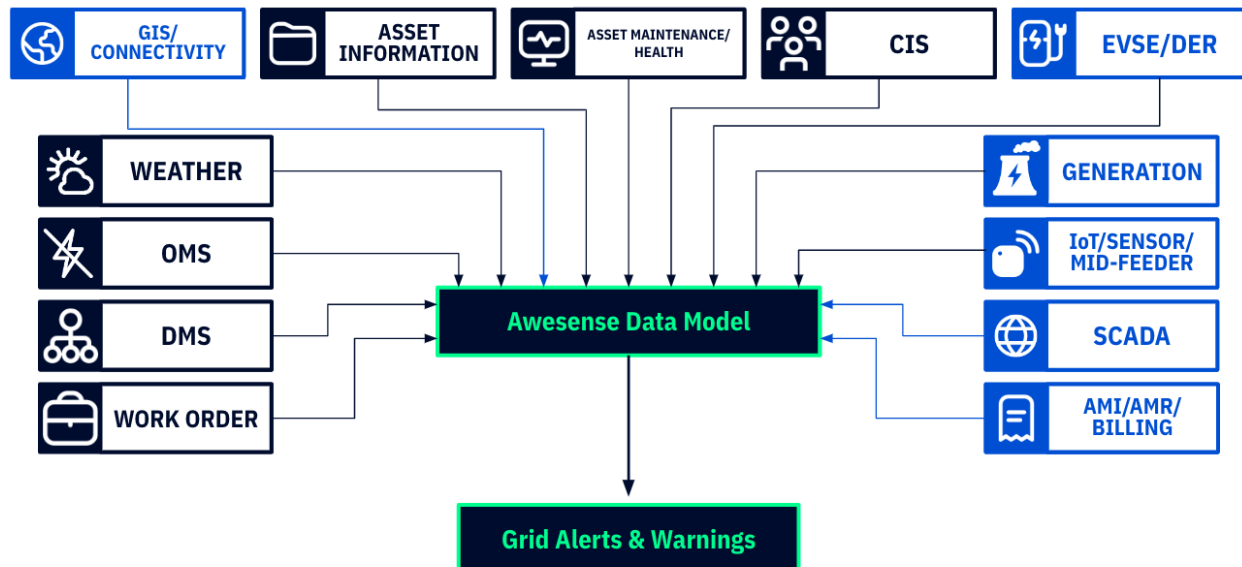
The Awesense Energy Transition Platform offers visualization using True Grid Intelligence (TGI) software, and all notifications can be displayed directly in this interface. TGI includes a Grid Alerts page where alerts are displayed:

- In the tabular form where all attributes of the warning or alert are listed. Additional information includes data as such element ID or the phase where the threshold has been exceeded;
- On a map, where the size of the circles on the map indicates the number of alerts at a location.

Awesense's TGI offers an easy, manageable and systematic approach to setting the grid alerts and warnings to detect the issues in the distribution grid immediately as they happen.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Grid Alerts - Please speak to Awesense about access to the sandbox environment or if you would like to see a demo of the solution.